

Предварительная настройка оборудования Cisco

Подключение, базовая конфигурация и удалённый доступ

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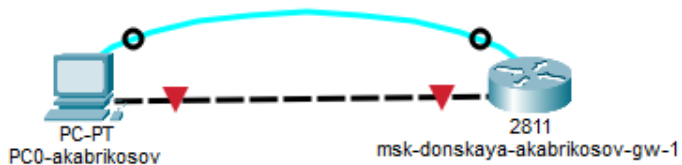
15 февраля 2026

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Цели и задачи работы

Получить основные навыки по начальному конфигурированию сетевого оборудования Cisco и настройке удалённого доступа.

Построение сети



Назначение IP-адресов на ПК

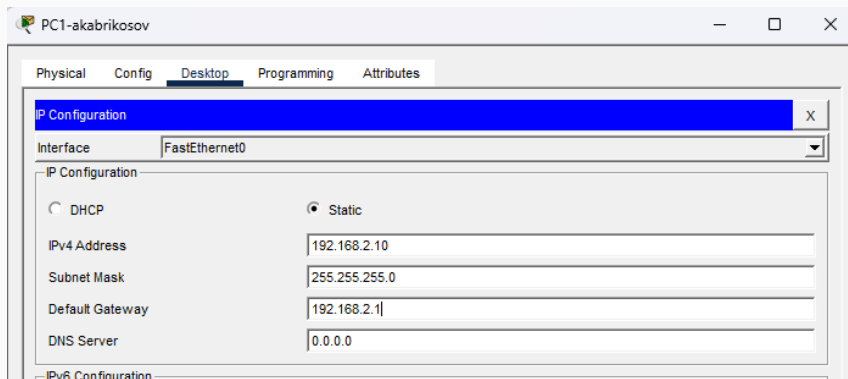


Рис. 2: Настройка IP-адреса на PC1

Назначение IP-адресов на ПК

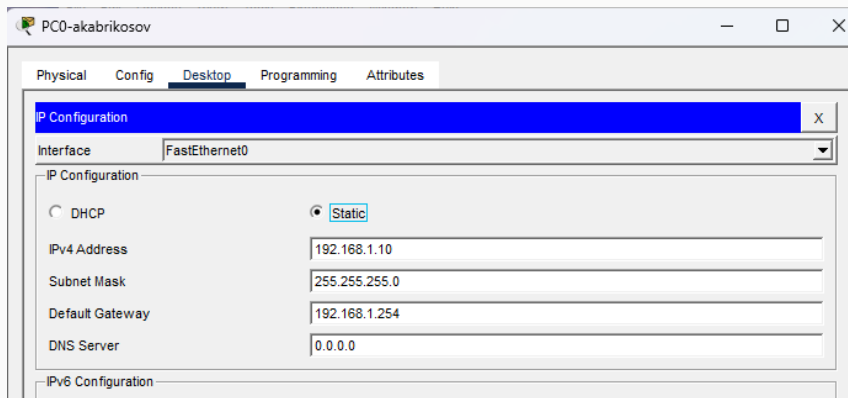
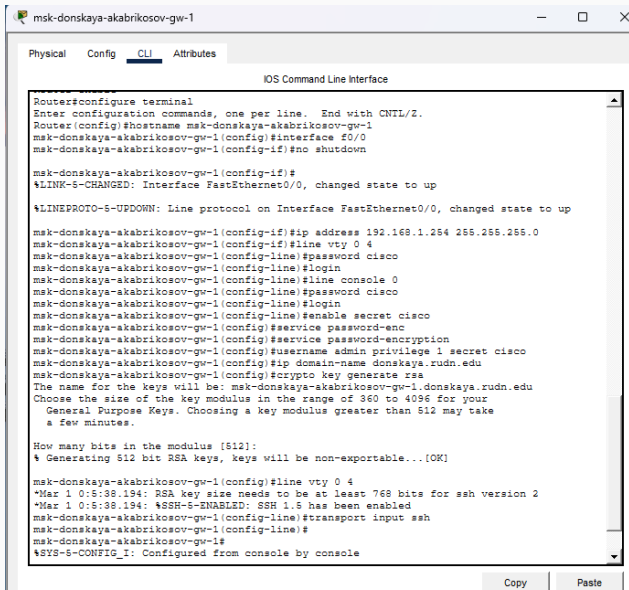


Рис. 3: Настройка IP-адреса на PC0

Настройка маршрутизатора



msk-donskaya-akabrikov-gw-1

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-akabrikov-gw-1
msk-donskaya-akabrikov-gw-1(config)#interface f0/0
msk-donskaya-akabrikov-gw-1(config-if)#no shutdown

msk-donskaya-akabrikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

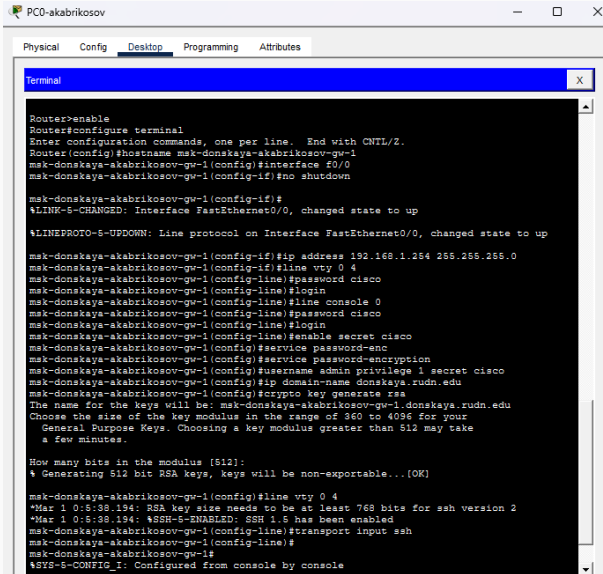
msk-donskaya-akabrikov-gw-1(config-if)#ip address 192.168.1.254 255.255.255.0
msk-donskaya-akabrikov-gw-1(config-if)#line vty 0 4
msk-donskaya-akabrikov-gw-1(config-line)#password cisco
msk-donskaya-akabrikov-gw-1(config-line)#login
msk-donskaya-akabrikov-gw-1(config-line)#line console 0
msk-donskaya-akabrikov-gw-1(config-line)#password cisco
msk-donskaya-akabrikov-gw-1(config-line)#login
msk-donskaya-akabrikov-gw-1(config-line)#enable secret cisco
msk-donskaya-akabrikov-gw-1(config)#service password-enc
msk-donskaya-akabrikov-gw-1(config)#service password-encryption
msk-donskaya-akabrikov-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-akabrikov-gw-1(config)#ip domain-name donskey.rudn.edu
msk-donskaya-akabrikov-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-akabrikov-gw-1.donskey.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-akabrikov-gw-1(config)#line vty 0 4
*Mar 1 0:5:38.194: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:5:38.194: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-akabrikov-gw-1(config-line)#transport input ssh
msk-donskaya-akabrikov-gw-1(config-line)#
msk-donskaya-akabrikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Включение SSH-доступа



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-akabrikosov-gw-1
msk-donskaya-akabrikosov-gw-1(config)#interface f0/0
msk-donskaya-akabrikosov-gw-1(config-if)#no shutdown

msk-donskaya-akabrikosov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

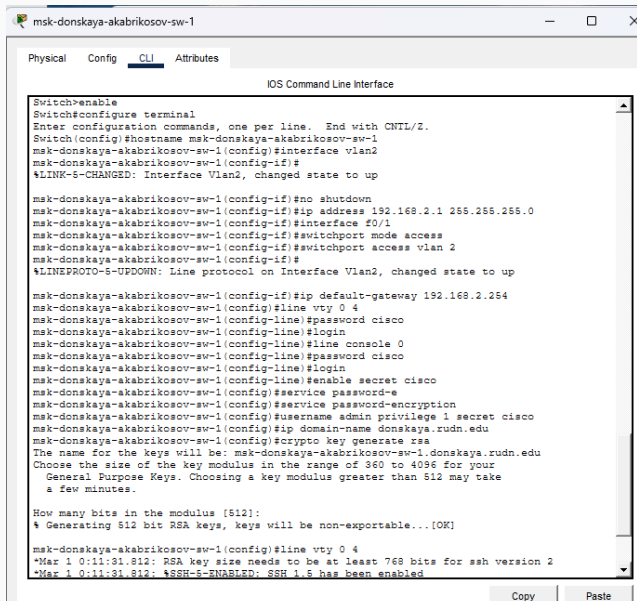
msk-donskaya-akabrikosov-gw-1(config-if)#ip address 192.168.1.254 255.255.255.0
msk-donskaya-akabrikosov-gw-1(config-if)#line vty 0 4
msk-donskaya-akabrikosov-gw-1(config-line)#password cisco
msk-donskaya-akabrikosov-gw-1(config-line)#login
msk-donskaya-akabrikosov-gw-1(config-line)#line console 0
msk-donskaya-akabrikosov-gw-1(config-line)#password cisco
msk-donskaya-akabrikosov-gw-1(config-line)#login
msk-donskaya-akabrikosov-gw-1(config-line)#enable secret cisco
msk-donskaya-akabrikosov-gw-1(config)#service password-enc
msk-donskaya-akabrikosov-gw-1(config)#service password-encryption
msk-donskaya-akabrikosov-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-akabrikosov-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-akabrikosov-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-akabrikosov-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-akabrikosov-gw-1(config)#line vty 0 4
*Mar 1 0:5:38.194: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:5:38.194: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-akabrikosov-gw-1(config-line)#transport input ssh
msk-donskaya-akabrikosov-gw-1(config-line)#
msk-donskaya-akabrikosov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Настройка коммутатора

Настройка интерфейса управления



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-akabrikosov-sw-1
msk-donskaya-akabrikosov-sw-1(config)#interface vlan2
msk-donskaya-akabrikosov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

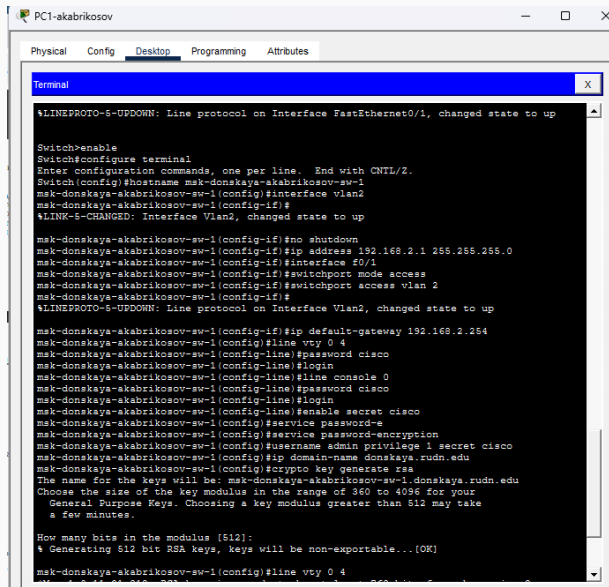
msk-donskaya-akabrikosov-sw-1(config-if)#no shutdown
msk-donskaya-akabrikosov-sw-1(config-if)#ip address 192.168.2.1 255.255.255.0
msk-donskaya-akabrikosov-sw-1(config-if)#interface #0/1
msk-donskaya-akabrikosov-sw-1(config-if)#switchport mode access
msk-donskaya-akabrikosov-sw-1(config-if)#switchport access vlan 2
msk-donskaya-akabrikosov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-akabrikosov-sw-1(config-if)#ip default-gateway 192.168.2.254
msk-donskaya-akabrikosov-sw-1(config)#line vty 0 4
msk-donskaya-akabrikosov-sw-1(config-line)#password cisco
msk-donskaya-akabrikosov-sw-1(config-line)#login
msk-donskaya-akabrikosov-sw-1(config-line)#line console 0
msk-donskaya-akabrikosov-sw-1(config-line)#password cisco
msk-donskaya-akabrikosov-sw-1(config-line)#login
msk-donskaya-akabrikosov-sw-1(config-line)#enable secret cisco
msk-donskaya-akabrikosov-sw-1(config)#service password-e
msk-donskaya-akabrikosov-sw-1(config)#service password-encryption
msk-donskaya-akabrikosov-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-akabrikosov-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-akabrikosov-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-akabrikosov-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-akabrikosov-sw-1(config)#line vty 0 4
*Mar 1 0:11:31.812: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:11:31.812: %SSH-5-ENABLED: SSH 1.5 has been enabled
```

Настройка портов и SSH



```
PC1-akabrikosov
Physical Config Desktop Programming Attributes
Terminal
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-akabrikosov-sw-1
msk-donskaya-akabrikosov-sw-1(config)#interface vlan2
msk-donskaya-akabrikosov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-akabrikosov-sw-1(config-if)#no shutdown
msk-donskaya-akabrikosov-sw-1(config-if)#ip address 192.168.2.1 255.255.255.0
msk-donskaya-akabrikosov-sw-1(config-if)#interface f0/1
msk-donskaya-akabrikosov-sw-1(config-if)#switchport mode access
msk-donskaya-akabrikosov-sw-1(config-if)#switchport access vlan 2
msk-donskaya-akabrikosov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

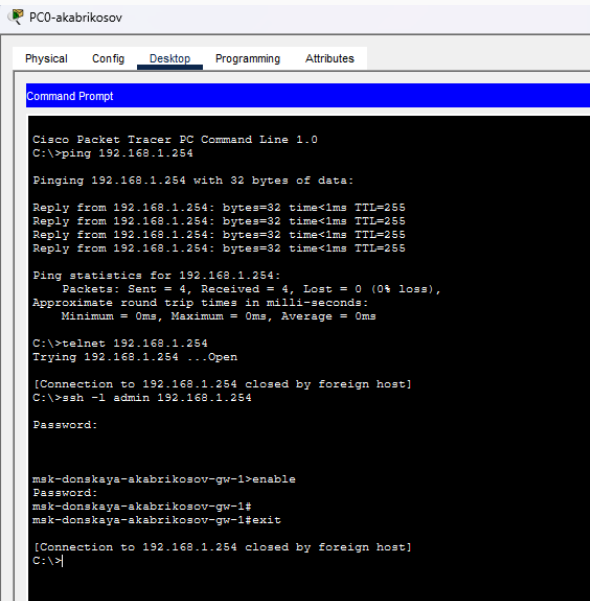
msk-donskaya-akabrikosov-sw-1(config-if)#ip default-gateway 192.168.2.254
msk-donskaya-akabrikosov-sw-1(config)#line vty 0 4
msk-donskaya-akabrikosov-sw-1(config-line)#password cisco
msk-donskaya-akabrikosov-sw-1(config-line)#login
msk-donskaya-akabrikosov-sw-1(config-line)#line console 0
msk-donskaya-akabrikosov-sw-1(config-line)#password cisco
msk-donskaya-akabrikosov-sw-1(config-line)#login
msk-donskaya-akabrikosov-sw-1(config-line)#enable secret cisco
msk-donskaya-akabrikosov-sw-1(config)#service password-e
msk-donskaya-akabrikosov-sw-1(config)#service password-encryption
msk-donskaya-akabrikosov-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-akabrikosov-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-akabrikosov-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-akabrikosov-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-akabrikosov-sw-1(config)#line vty 0 4
```

Проверка работоспособности сети

Проверка связи с маршрутизатором



The screenshot shows the Cisco Packet Tracer PC Command Line interface for a PC named 'PC0-akabrikosov'. The interface has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.1.254
Trying 192.168.1.254 ...Open

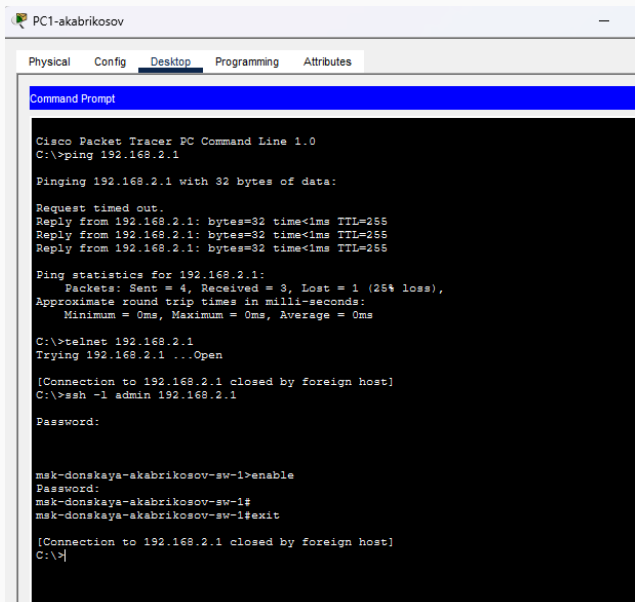
[Connection to 192.168.1.254 closed by foreign host]
C:\>ssh -l admin 192.168.1.254

Password:

msk-donskaya-akabrikosov-gw-1>enable
Password:
msk-donskaya-akabrikosov-gw-1#
msk-donskaya-akabrikosov-gw-1#exit

[Connection to 192.168.1.254 closed by foreign host]
C:\>
```

Проверка связи с коммутатором



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'PC1-akabrikosov'. The 'Desktop' tab is selected. The command prompt shows the following sequence of commands and outputs:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.2.1
Trying 192.168.2.1 ...Open

[Connection to 192.168.2.1 closed by foreign host]
C:\>ssh -l admin 192.168.2.1

Password:

msk-donskaya-akabrikosov-sw-1>enable
Password:
msk-donskaya-akabrikosov-sw-1#
msk-donskaya-akabrikosov-sw-1#exit

[Connection to 192.168.2.1 closed by foreign host]
C:\>|
```


Выводы по проделанной работе

В ходе выполнения лабораторной работы была построена сеть в Cisco Packet Tracer с использованием маршрутизатора, коммутатора и двух конечных устройств.

Выполнена базовая настройка сетевого оборудования, назначены IP-адреса, настроены параметры безопасности и организован защищённый удалённый доступ по протоколу SSH.

Работоспособность сети подтверждена с помощью проверки доступности устройств и успешного подключения к ним.